Accelerating the Discovery of Complex Biomolecular States with Enhanced Sampling

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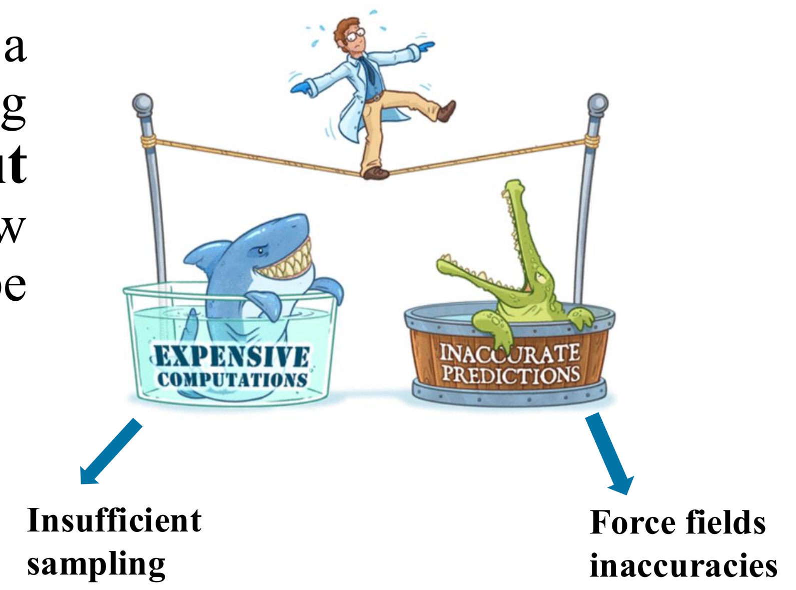
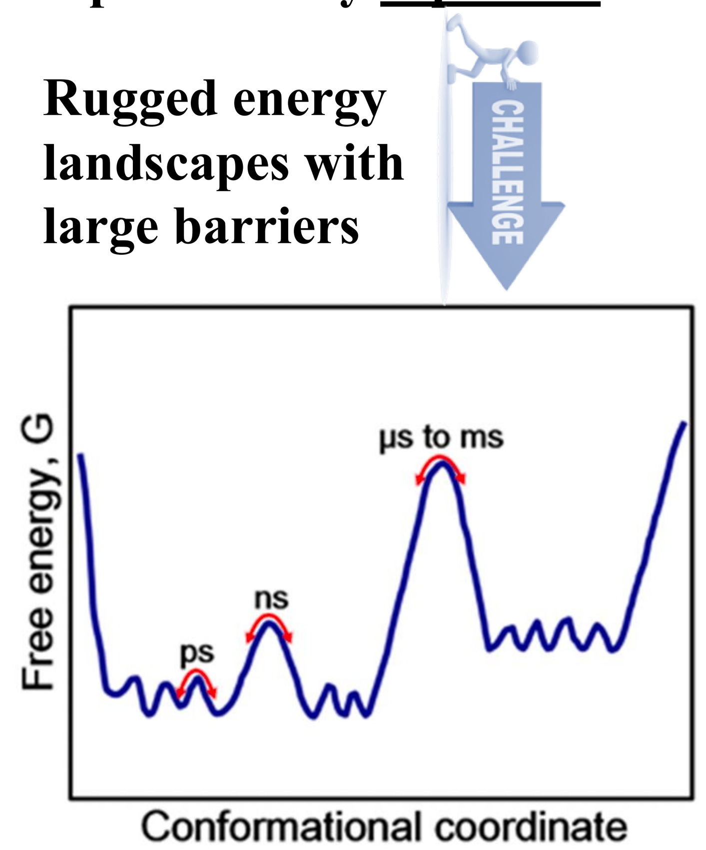
Background and Motivation

>> **Molecular dynamics (MD)** is a powerful tool for studying biomolecular dynamics, **but <u>accurate</u>** sampling of slow conformational changes can be **computationally <u>expensive</u>**.



Adapted from Wolf et. al. *J. Chem. Educ.* 2022, 99, 3, 1479–1487

Rugged energy landscapes with large barriers



Henzler-Wildman, K., & Kern, D. (2007). *Nature*, *450*(7172), 964-972.

>> GPU-accelerated MD helps, **but** an unfeasible amount of computational time might be needed to sample all the relevant metastable states.

>> **Enhanced Sampling** helps **facilitate the crossing of energy barriers**.

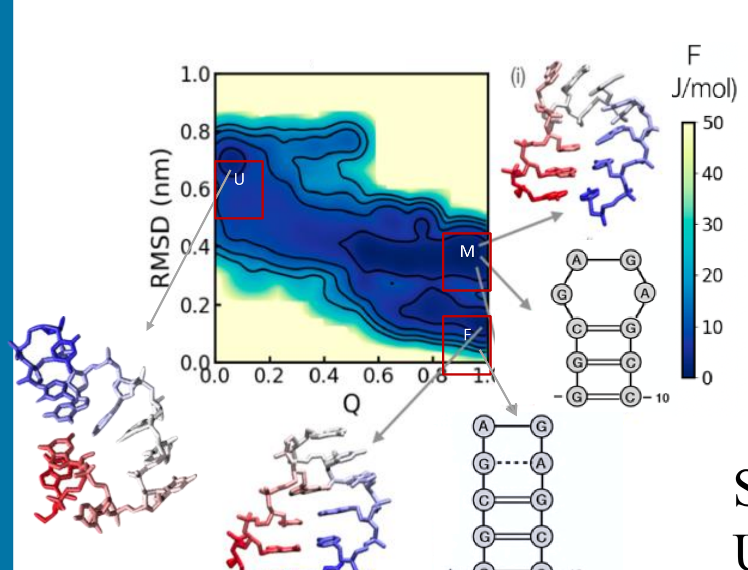
Collective variable (CV)-based sampling	Tempering-based sampling
Not all degrees of freedom may be captured by CVs	Limited barrier crossing

>> **Combining tempering-based and collective variable-based enhanced sampling techniques** to overcome limitations

Adapting MM-OPES for atomistic MD simulations of biomolecules

>> Conventional enhanced sampling techniques are **successful but still require substantial GPU resources.**

PTWTE-WTM: Parallel-Tempering in the Well-Tempered Ensemble combined with Well-Tempered Metadynamics



Zerze, G. H., et al. *The Journal of Physical Chemistry B*, *125*.50 (2021), 13685-13695.

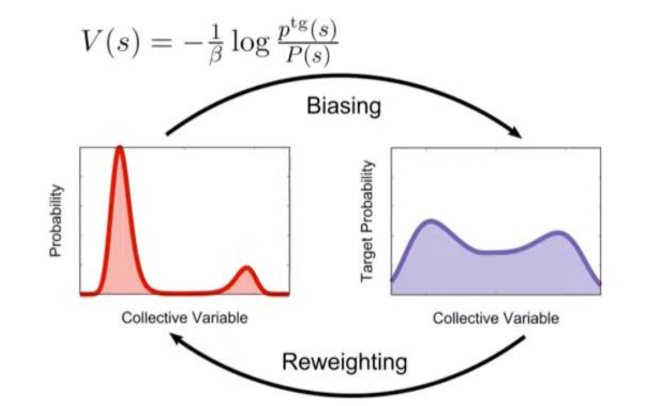
>> MD engine: GROMACS [2020.5]
>> Enhanced sampling: PLUMED [2.8]
>> GPU model: Volta V100
>> Force fields: DESRES ff and TIP4PD
>> 14 replicas,
ranging between 300 and 475 K

Successfully samples metastable states: Unfolded (**U**), Folded (**F**), Misfolded (**M**)

>> Can the **M**ultithermal-**M**ultiumbrella **O**n-the-fly **P**robability **E**nhanced **S**ampling (**MM-OPES**) **reliably <u>replace</u>** the PTWTE-WTM and be more **<u>efficient</u>**?

Results and Discussion

OPES

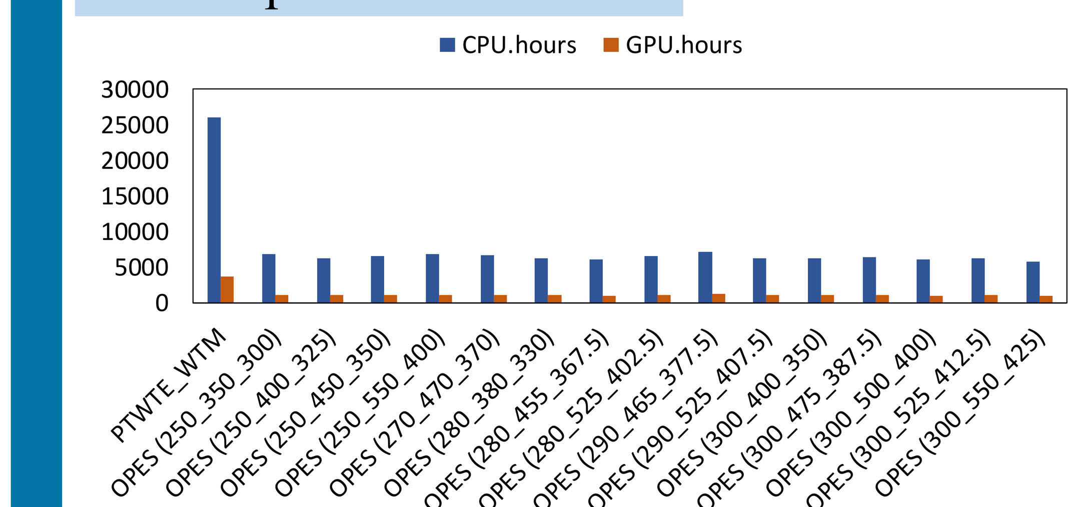


Invernizzi, M., & Parrinello, M.,2020, *The Journal of Physical Chemistry A*, *11*(7), 2731.

>> Taking PTWTE-WTM as a reference
>> 8 walkers MM-OPES simulations
>> Testing 15 temperature ranges to find optimal parameters and evaluating efficiency

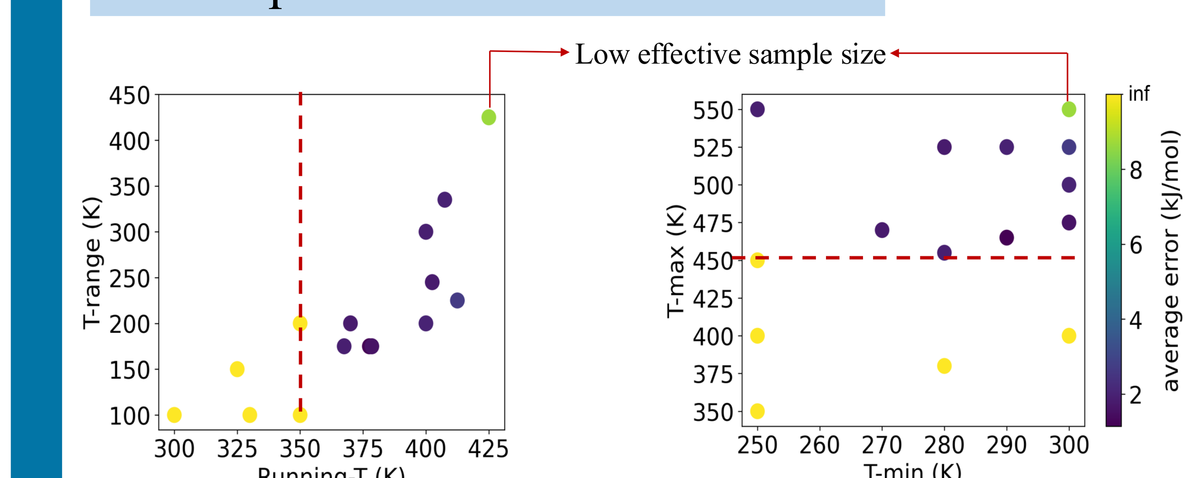
min T	max T	run T	min T	max T	run T
250	350	300	290	465	377.5
250	400	325	290	525	407.5
250	450	350	300	400	350
250	550	400	300	475	387.5
270	470	370	300	500	400
280	380	330	300	525	412.5
280	455	367.5	300	550	425
280	525	402.5			

>> Computational cost



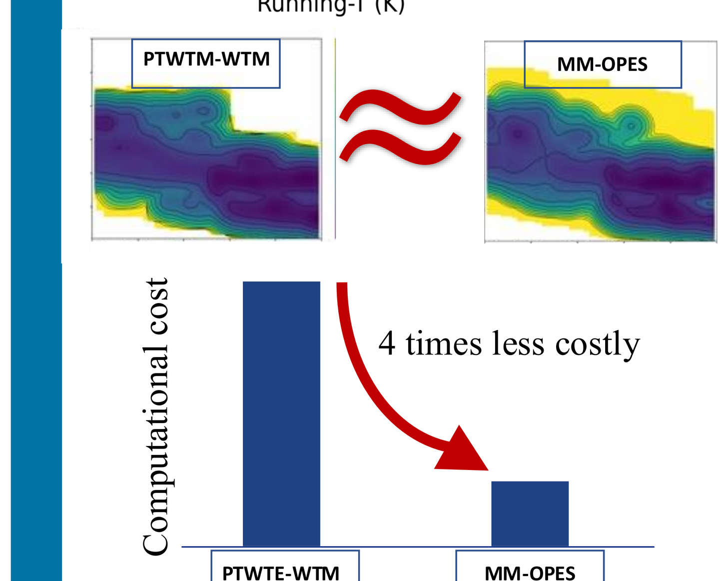
MM-OPES simulations employed ~4× fewer CPU and GPU hours than PTWTE-WTM.

>> Temperature-related variables



Conditions

- High Maximum T
- High Running T
- Reasonable effective sample size



More science per GPU-hour

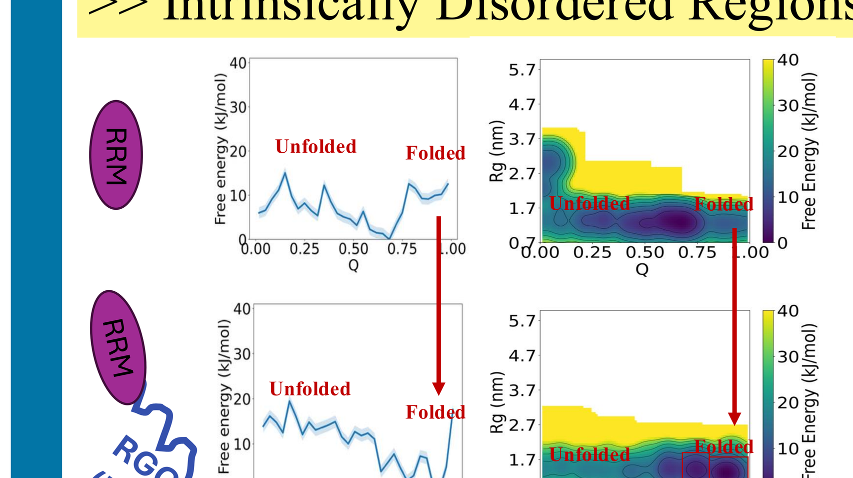
→ enabling atomistic investigation of **larger biomolecular systems** that are **difficult to characterize experimentally**

Rahimi, K., Piaggi, P.M. and Zerze, G.H., 2023. *The Journal of Physical Chemistry B*, 127(21), pp.4722-4732.

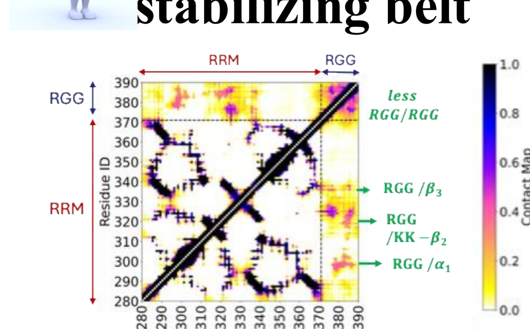
Extending MM-OPES to more complex biomolecular systems

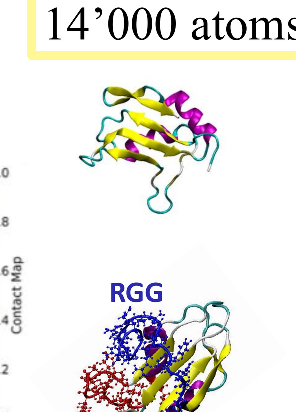
>> Intrinsically Disordered Regions (IDRs) flanking folded domains

14’000 atoms



IDR acts as a **stabilizing belt**

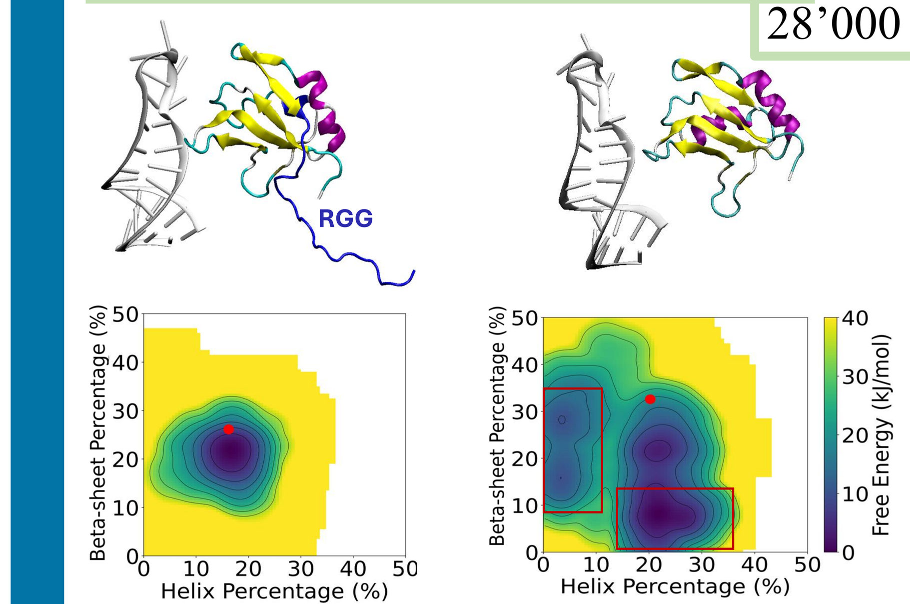




Rahimi, Kosar, et al. *The Journal of Physical Chemistry B* 130.27 (2026): 6773-6781.

>> Protein–RNA interactions

28’000 atoms



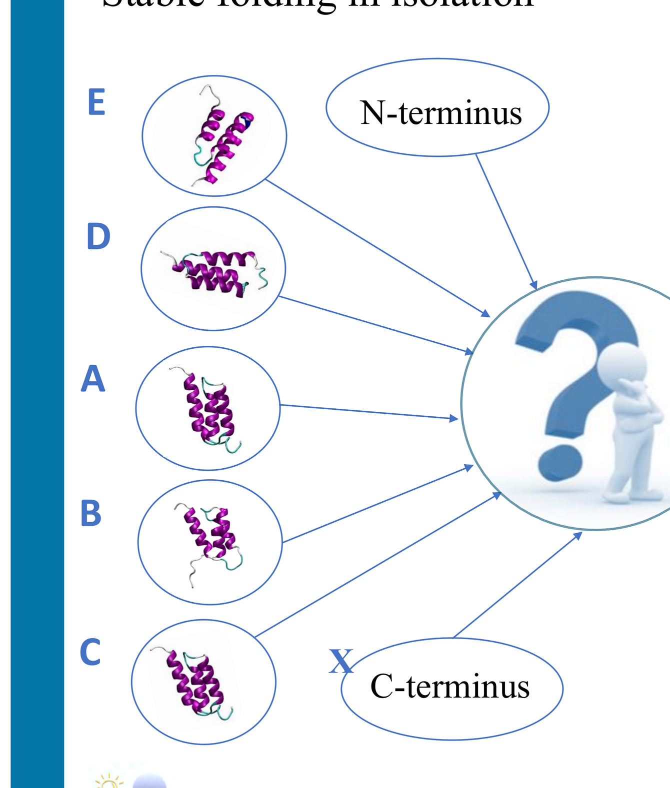
>> Does RNA proximity affect the stability of a protein?

RNA/RRM interactions make the **unfolded state more structured yet heterogeneous** in the absence of IDR.

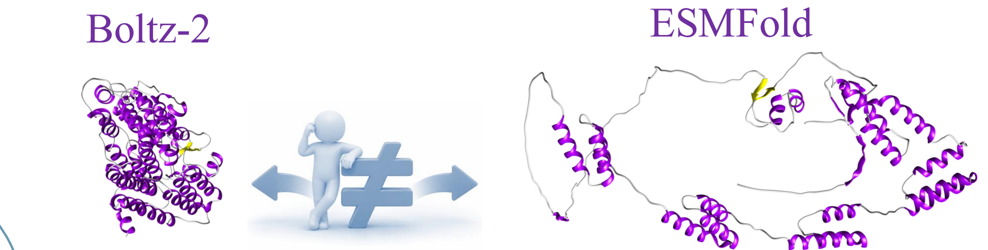
>> Multidomain proteins

180’000 atoms

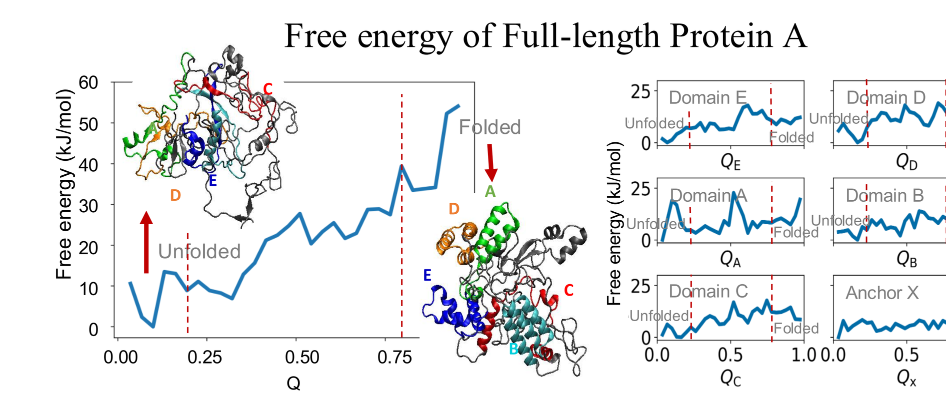
Stable folding in isolation

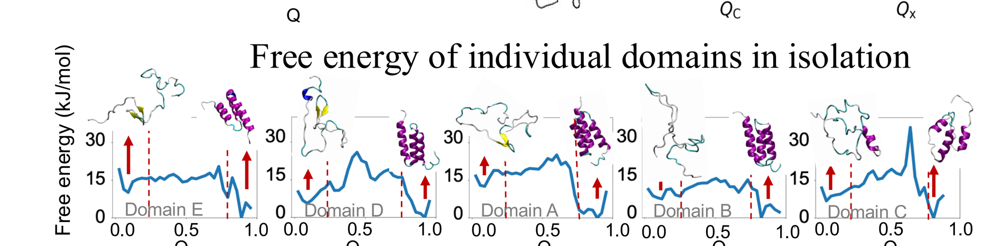


AI-based predictions for full-length Protein A



>> First atomistic simulation of full-length Protein A using MM-OPES



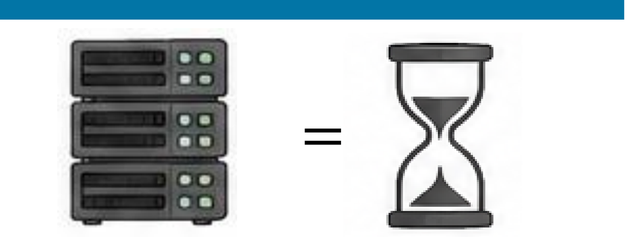


Protein A exhibits regional helicity rather than fully folded domains due to **interdomain contacts not captured by AI models.**

Rahimi, K., Halbing, A., Nguyen, M.N., Sen, M., Willson, R.C. and Zerze, G.H., 2025. *bioRxiv*.

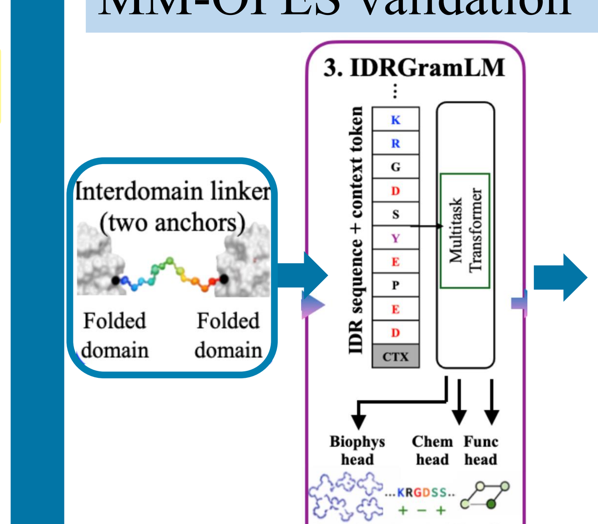
Conclusion

>> By **reducing computational cost in GPU-accelerated simulations**, MM-OPES enables atomistic investigation of complex biomolecular systems.



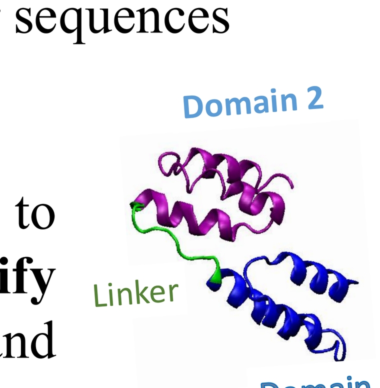
Future work

>> AI-guided IDR design for multidomain protein design with MM-OPES validation



Use AI to propose ideal linker sequences

Apply MM-OPES to selected candidates to **identify unexpected contacts** and assess conformational stability.



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